Arya Mukherjee

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Education

University of Bath *Sept 2019 - July 2024*

- MEng (Hons) Mechanical Engineering
- 2:1 Upper Second-Class Honours

Littleover Community School *2014 - 2019*

- A-Levels - Mathematics: A*, Physics: A, Computer Science: A
- 10 A* to As in GCSEs

Experience

Associate Software Engineer | SITA *Sep 2024 - Present*

- Designed and developed a distributed Python system for large-scale codebase analysis and summarisation, greatly reducing the onboarding time for new team members across the company - improving time-to-productivity by over 2 weeks.
- Architected and implemented a task parallelisation scheme for the summariser, distributing compute across 10 systems, improving inference speed by 500% and reducing monthly operational costs by 60%.
- Led an epic to modernise a partner team's testing stack by building a full stack migration tool from Java Selenium to Python Playwright, improving test maintainability and accelerating future test development.
- Written unit, regression, and end-to-end test suites in Python with pytest, and currently building an automated soak testing tool with Playwright to validate system stability under sustained load.
- Team lead and Scrum Master for 8 graduate developers, overseeing work delegation, mentoring and team coordination.
- **Technical Skills:** *Python, Docker, Linux, Git, Neo4j, Kubernetes, Playwright, pytest, uv, AI Foundry, Ollama, RAG, Azure, Agile, Tree-sitter*

Mechanical Engineer | PRAX Lindsey Oil Refinery *July 2021 - July 2022*

- Assisted in scheduling a £40m project, processing contractor data in Oracle Primavera to ensure accurate and timely project execution.
- Automated QA data workflows with Python and Pandas, reducing spreadsheet processing times by 2000%, improving reporting speed and reliability for key safety and compliance checks.
- Processed site-wide operational data into interactive dashboards, using SAP and PowerBI.
- **Technical Skills:** *Python, Pandas, NumPy, Re, PowerBI, PowerQuery, Oracle Primavera, SAP*

Projects

Physics Informed Neural Network Modeller (Master's Thesis) *Jan 2024 - Apr 2024*

- Developed a Physics Informed Neural Network in PyTorch to model 3D transient heat system PDEs in gas turbines, achieving highly accurate simulations which generated 10x faster than traditional numerical methods.
- Accelerated and improved training by over 100x with CUDA parallelisation, the L-BFGS optimiser, gradient clipping and mixed precision training.
- Conducted various network hyperparameter space studies to determine the optimal architecture, activation function and weight initialisations - to expedite network convergence whilst avoiding the vanishing gradient problem.
- **Tools used:** *Deep Learning, Neural Networks, PyTorch, PDEs, NVIDIA CUDA, Plotly, Code Profiling, Jupyter Notebooks*

Neural Network Options Pricer (In Progress) *Apr 2026 - Present*

- Scoping and designing a low-latency C++ options pricer using the Black–Scholes model, with a focus on concurrent architecture and reliability. The pricer will use a Physics Informed Neural Network, with a custom-built C++ neural network library.
- **Tools used:** *C++ 23, Multithreading, Black–Scholes-Merton, Backtesting*

Various Numerical Modelling and Simulation Projects *Sep 2020 - Dec 2023*

- Created multiple numerical solvers in MATLAB for partial differential equations in heat transfer, orbital mechanics and drug absorption.
- Utilised computer vision to extract graph data from various NASA publications, to model 3D heat profiles of a space shuttle upon re-entry.
- Applied vectorisation and multi-threading to accelerate calculations and improve efficiency.
- Enhanced interactivity by creating GUIs for the numerical solvers, allowing users to run simulations and visualise results without editing code.
- **Tools used:** *MATLAB, Partial Differential Equations, Computer Vision, Multi-Threading, OOP.*

Discord Bot *2022 - Present*

- Built and deployed an async event-driven Node.js application serving 1000+ users, containerised with Docker and backed by a MySQL database.
- Implemented fuzzy matching and string tokenization algorithms to improve real-time query processing, and built REST API integrations, interactive GUI components and user management tooling.
- **Tools used:** *Node.js, Docker, MySQL, Discord.js, REST APIs, Regex, String Tokenization*

Skills

Languages	Python, C/C++ (23), JavaScript/Node.js, SQL, MATLAB
Quantitative	PDEs, Numerical Methods, Linear Algebra, Black–Scholes, Neural Networks, Optimisation
Libraries	Playwright, pytest, py-spy, NumPy, PyTorch, Pandas, CUDA, Plotly
Infrastructure	Docker, Kubernetes, Linux, Git, Azure, MySQL, Neo4j, uv, PM2, Shell (Bash/Fish)
Tools	Jupyter Notebooks, PowerBI, Ollama, Tree-sitter, Scalene Profiling, LaTeX